Final Project Report

Employee Raise Issue – Record Producer & Service Portal Widget

ServiceNow Global Certification Program

1. INTRODUCTION

1.1 Project Overview

In a large enterprise organization, employees frequently face workplace-related issues such as payroll discrepancies, access problems, infrastructure requests, or HR policy clarifications. These issues are currently raised through emails or informal communication channels, leading to a lack of standardized data capture, delayed resolution, no visibility into issue status, manual effort for HR and support teams, and inconsistent prioritization and assignment.

This project configures a Record Producer for a custom “Employee Raise Issue” table and integrates it with a Service Portal widget, allowing employees to raise issues easily from the Employee Center Service Portal while enforcing business rules and dependencies.

1.2 Purpose

The purpose of this project is to replace unstructured, email-based issue reporting with a guided, structured form that captures consistent data, automatically routes and prioritizes issues to the right HR/IT group, and gives employees visibility into the status of what they've raised — reducing manual triage effort and improving resolution time.

2. IDEATION PHASE

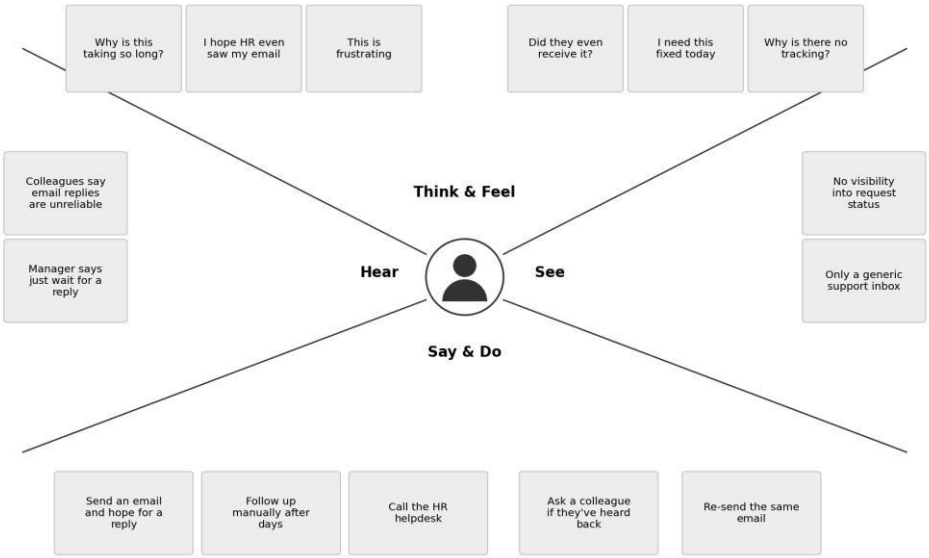
2.1 Problem Statement

Two customer problem statements were defined to capture the perspective of the employee raising an issue and the HR/IT agent handling it:

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	An employee	raise a workplace issue (payroll, access, infrastructure or HR policy) and know it will be looked at	there is no standard form and no way to track its status after sending an email	issues are logged informally over email/chat with no ticketing, categorization or ownership	frustrated and unheard, unsure whether the issue is even being worked on
PS-2	An HR / IT support agent	receive, prioritize and resolve employee issues efficiently	requests arrive unstructured, are not categorized or assigned automatically, and pile up in a shared inbox	there is no dedicated table, record producer or portal widget to capture and route issues consistently	overwhelmed and unable to prioritize, risking delayed or missed resolutions

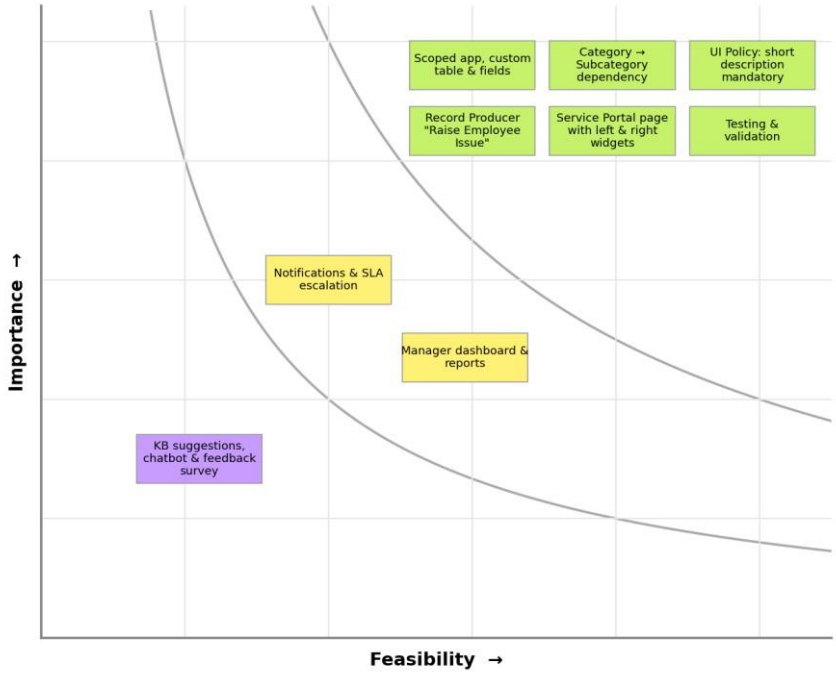
2.2 Empathy Map Canvas

An empathy map was built for the employee persona to understand their thoughts, feelings, and behaviour when raising an issue through the old, unstructured process:



2.3 Brainstorming

The team brainstormed ideas addressing the problem statement and prioritized them on an importance vs. feasibility grid. High-priority (“Do first”) ideas — the scoped app and table, dependency logic, UI Policy validation, the Record Producer, the Service Portal page with widgets, and testing — were selected for this release; notifications, SLA/escalation and dashboards were planned as next steps, with Knowledge Base suggestions and chatbot entry parked for later:



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The employee's journey was mapped across five phases — Entice, Enter, Engage, Exit, and Extend — covering steps, interactions, goals, positive/negative moments, and opportunities for improvement:



3.2 Solution Requirement Functional Requirements:

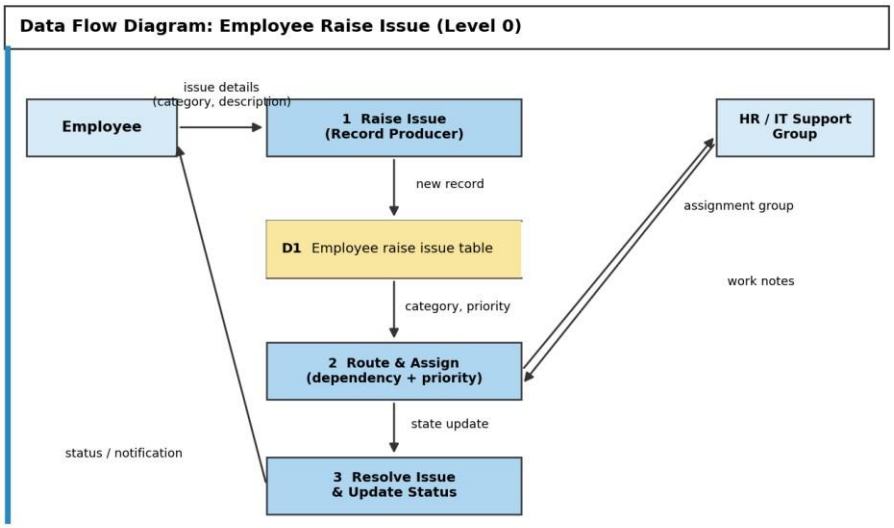
FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	Issue Submission	• Submit an issue via the Employee Raise Issue Record Producer on the Service Portal • Attach supporting files or screenshots to an issue
FR-2	Field Validation & Dependency	• Category → Subcategory dependent choice list • Mandatory short description enforced via UI Policy
FR-3	Issue Routing & Assignment	• Auto-assign new issues to an HR/IT group based on category • Auto-calculate priority from impact and urgency
FR-4	Issue Tracking & Notification	• View issue status via a “My Issues” widget on the portal • Notify the employee by email when the issue state changes

Non-Functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Record Producer and Service Portal widgets should be simple and guided enough to use without training.
NFR-2	Security	Role-based ACLs ensure employees can view only their own issues, while HR/IT see records for their assignment group.
NFR-3	Reliability	Issue submissions and state changes should not fail or lose data, and notifications should be delivered consistently.
NFR-4	Performance	The portal page, Record Producer and dependent dropdowns should respond within a few seconds under normal load.
NFR-5	Availability	The Service Portal and Record Producer should remain accessible during business hours with minimal downtime.
NFR-6	Scalability	The table, workflow and assignment logic should handle a growing number of employees and issues without redesign.

3.3 Data Flow Diagram

The Level-0 data flow diagram shows how an issue moves from the employee through the Record Producer, into the Employee raise issue table, through routing/assignment, to the HR/IT group and back to the employee as a status update:



3.4 Technology Stack

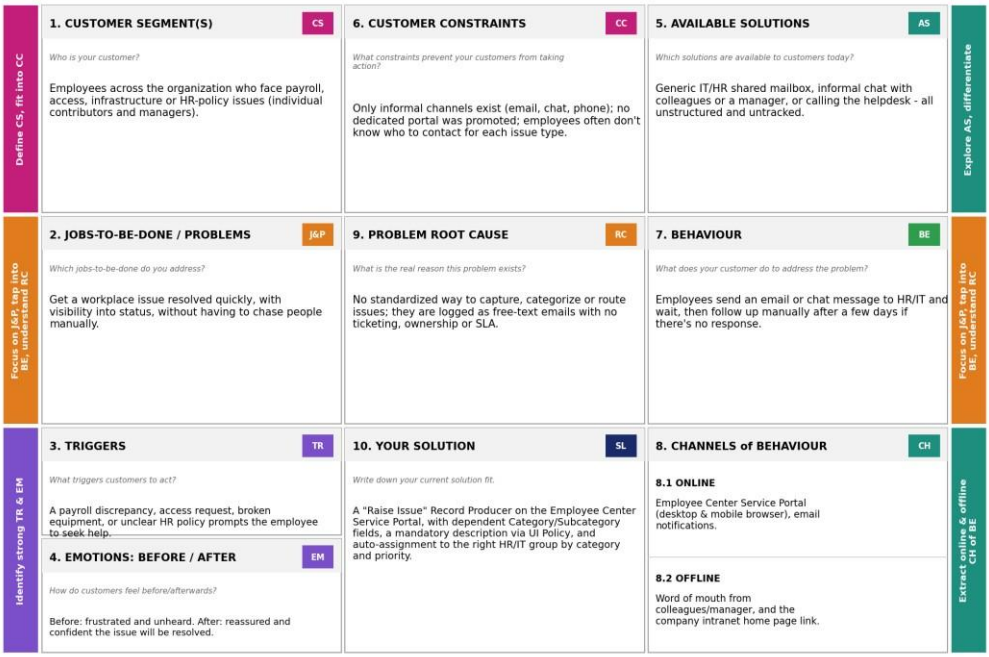
The solution is built entirely on the ServiceNow Now Platform:

S.No	Component	Description	Technology
1.	User Interface	Employee Center Service Portal page and widgets used to raise and view issues	HTML, CSS, AngularJS (Service Portal widget framework), Bootstrap
2.	Application Logic	Record Producer logic, dependency, validation, auto-assignment & priority	Glide (server-side JS), Client Scripts & UI Policies
3.	Database	Storage for the custom “Employee raise issue” table	ServiceNow platform database (Now Platform)
4.	File Storage	Storage for attachments added to an issue	ServiceNow Attachment API
5.	Notifications	State-change notifications to employees	ServiceNow Email/Notification API
6.	Infrastructure	Application deployment	Now Platform SaaS instance (ServiceNow Cloud)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit canvas maps the customer segment, constraints, available (old) solutions, jobs-to-be-done, root cause, behaviour, triggers, emotions, the proposed solution, and channels of behaviour:

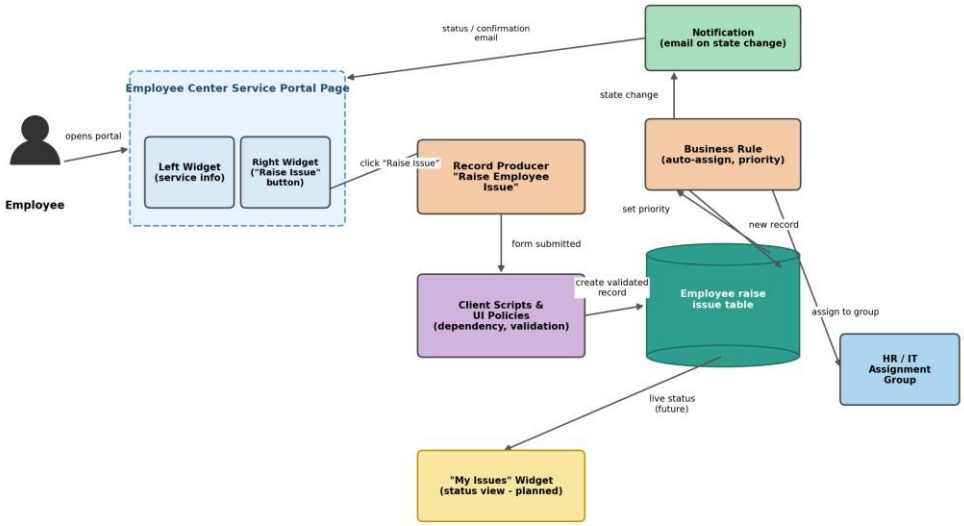


4.2 Proposed Solution

S.No.	Parameter	Description
1.	**Problem Statement (Problem to be solved)**	Employees raise workplace issues through email or informal channels, causing delayed resolution, no visibility into status, inconsistent prioritization, and manual effort for HR and support teams.
2.	**Idea / Solution description**	Configure a Record Producer for a custom “Employee Raise Issue” table and integrate it with a Service Portal widget, so employees can raise issues from a guided form while issues are auto-assigned and prioritized.
3.	**Novelty / Uniqueness**	Combines a guided, dependency-driven form with automatic routing and priority calculation, purpose-built around this organization's actual issue types rather than a generic ticket form.
4.	**Social Impact / Customer Satisfaction**	Employees feel heard and can track their issue; HR/IT spend less time manually triaging a shared inbox.
5.	**Business Model (Revenue Model)**	Internal enterprise tool with no external revenue model; value is delivered as reduced manual handling cost and improved employee satisfaction.
6.	**Scalability of the Solution**	Built as a scoped application on a configurable table, so new categories, groups or departments can be added without redesigning the workflow.

4.3 Solution Architecture

The solution architecture shows how the Employee Center Service Portal Page, Record Producer, Client Scripts/UI Policies, the Employee raise issue table, Business Rule, HR/IT Assignment Group, and Notification component fit together:



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The product backlog was broken into 10 user stories across 2 sprints, with story points assigned on a Fibonacci scale (1, 2, 3, 5):

Sprint	Epic	US #	User Story	Pts	Priority
Sprint-1	Issue Submission	USN-1	Raise a workplace issue via the Record Producer.	3	High
Sprint-1		USN-2	Dependent Category → Subcategory choice list.	2	High
Sprint-1		USN-3	Mandatory short description before submit.	1	Medium
Sprint-2		USN-4	Attach a screenshot or document.	2	Low
Sprint-1	Issue Tracking	USN-5	Confirmation shown once issue is submitted.	1	High
Sprint-2		USN-6	View status of past issues (“My Issues” widget).	5	Medium
Sprint-1	Issue Routing	USN-7	Auto-assign issues to the correct group.	3	High
Sprint-2		USN-8	Priority based on impact and urgency.	2	Medium
Sprint-1	Issue Resolution	USN-9	Update issue state (New→In progress→Resolved→Closed).	2	High
Sprint-1	Configuration	USN-10	Configure dependency and UI Policies.	3	High

Sprint tracker:

Sprint	Total Points	Duration	Start Date	End Date (Planned)	Points Completed
Sprint-1	15	6 Days	03 Feb 2025	08 Feb 2025	15
Sprint-2	9	6 Days	10 Feb 2025	15 Feb 2025	9

Total story points = 15 + 9 = 24 across 2 sprints, giving the team a velocity of 12 story points per sprint (24 ÷ 2).

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional and performance test cases were designed to cover field validation, dependency logic, issue creation, auto-assignment, response time, and load:

Test ID	Scenario	Test Steps	Expected Result
FT-01	Category & Subcategory Field Validation	Select different categories and check the Subcategory list.	Subcategory list updates to show only relevant options.
FT-02	Mandatory Field Validation	Submit with short description left blank.	Submission blocked with a validation message.
FT-03	Issue Submission	Fill all required fields and click Submit.	Record created in the Employee raise issue table.
FT-04	Auto-Assignment & Priority	Submit issues under different categories.	Issue routed to the correct group with correct priority.
PT-01	Response Time	Time the Record Producer load and submit.	Should load and submit within 3 seconds.
PT-02	Concurrent Submission	Multiple employees submit at the same time.	Portal remains responsive; no records lost.
PT-03	Attachment Upload Load	Upload multiple/large files to an issue.	Upload completes without errors or crashing.

UAT execution against the deployed Record Producer surfaced 3 open defects, out of 8 test cases run (1 Pass, 3 Fail, 4 not yet tested):

Bug ID	Description	Severity	Status	Feedback
BG-001	Priority field remains “--None--” after saving a new issue record	High	Open	Priority should autocalculate from impact/urgency; currently left blank.
BG-002	Assignment Group is not populated on record creation	High	Open	Auto-assignment Business Rule does not appear to be setting the group.
BG-003	Subcategory value not restricted to the selected Category’s mapped options	Medium	Open	Record was saved with Category = Software, Subcategory = Laptop (belongs to Hardware).

7. RESULTS

7.1 Output Screenshots

The Record Producer “Raise Employee Issue” form, showing the Category/Subcategory fields and the resulting created record:

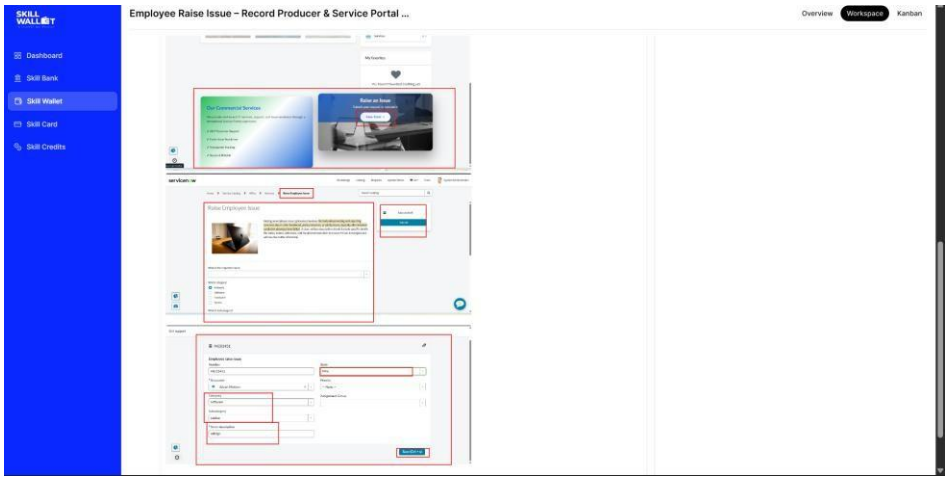


Figure: Service Portal – Raise Employee Issue form and the created record (INC01451)

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Standardizes issue capture through a single guided form instead of unstructured email or chat.
- Reduces manual triage effort for HR/IT through automatic assignment and priority calculation.
- Improves transparency — issues are categorized, timestamped, and trackable by state.
- Enforces data quality through Category→Subcategory dependency and mandatory field validation.
- Built natively on the existing ServiceNow platform, requiring no new infrastructure.

8.2 Disadvantages / Limitations

- No employee-facing “My Issues” status widget yet — planned but not built in this release.
- No automatic notification on state change yet.
- Priority and Assignment Group are not yet reliably auto-populated (see defects BG-001, BG-002).
- Subcategory dependency filtering has a known bug (BG-003).
- Limited to a single portal experience; no dedicated mobile app UI.

9. CONCLUSION

This project successfully replaced an informal, email-based process for raising workplace issues with a structured Record Producer and Service Portal widget on the ServiceNow platform. The core submission, dependency validation, and routing flow is functional, and UAT testing has surfaced three concrete defects (priority calculation, assignment group population, and subcategory dependency filtering) that need to be resolved before wider rollout. Once fixed, the solution is expected to meaningfully reduce manual HR/IT effort and give employees clearer visibility into their raised issues.

10. FUTURE SCOPE

- Build the “My Issues” widget for employees to self-serve check status.
- Add automatic email/in-app notifications on every issue state change.
- Resolve the open defects: priority calculation, assignment group population, subcategory dependency.
- Add SLA timers and automatic escalation for overdue issues.
- Build a manager/HR dashboard with ageing and volume reports.
- Add Knowledge Base suggestions and Virtual Agent deflection for self-service resolution.
- Extend the solution to additional departments or portals beyond the initial rollout.

11. APPENDIX

Source Code (if any): Configured natively within the ServiceNow instance (Record Producer, Business Rules, Client Scripts, UI Policies, and Service Portal widgets). Export as an Update Set if a portable source artifact is required.

Dataset Link: Not applicable — no external dataset is used in this solution.

GitHub & Project Demo Link:https://github.com/Varshitha-0412/Employee-Raise-Issue